Anuj Pratap Singh Yadav

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SUMMARY

Staff engineer with 11 years of C++, the last three-plus in formal verification at Synopsys. I own the coverage subsystems of a C++ vs RTL equivalence app end to end, lead architecture changes across team boundaries in the formal platform, and build LLVM-based analysis of C++ designs.

SKILLS

Formal verification: VC Formal (DPV: C++ vs RTL equivalence; FPV), formal coverage and property density, cone of influence, C++ and RTL code coverage, sign-off flows, distributed proofs

Languages: C++, Tcl, Python, C#, Java, TypeScript

Tools: Verdi, LLVM IR passes (dg), LSF, FlexNet, GDB, Linux, Perforce, Git, Win32

C++ architecture: dependency inversion, IWYU, serialization (XML, pugixml)

AI tooling: Cursor agent skills, rules and subagents built for the team's workflow, shared with the team

EXPERIENCE

R&D Staff Engineer | Synopsys *Noida, India | May 2023 - Present*

VC Formal (formal verification), DPV app: C++ vs RTL equivalence

- **C++ code coverage for C++ vs RTL sign-off**: own it end to end. It shows which parts of the C++ reference model the formal proofs actually exercised, so the C++ side can be signed off like RTL. Added merging across proofs (a block is covered if any proof covers it, unreachable only if every proof proves it) and export to the standard coverage database, so C++ and RTL coverage are reviewed together in Verdi; drive changes with the viewer and GUI teams, handle customer issues, and guided its integration into a new C++ + RTL sign-off app.
- **Formal coverage (property density, RTL coverage)**: own both. They report which design signals, down to the bit, and which RTL lines the properties actually check. Rebuilt property density so DPV and FPV give the same answer, adding bit-level results, RTL-to-RTL designs and a PASS/FAIL output-coverage check (beta); aligned RTL coverage semantics across the two apps. In production.
- **C++ cone of influence (R&D)**: building source-level cone of influence for C++ designs, which reports the exact C++ lines a property depends on. Wrote LLVM IR passes on the open-source dg slicer for C++ constructs it missed, and a signal-based slicing criterion that cut whole-file slices down to the lines feeding each signal. Design reviewed weekly with a Distinguished Engineer.
- **Build-dependency decoupling (multi-org initiative)**: led VC Formal's part. The shared C++ layer under every formal app included engine headers, so one engine edit rebuilt every app. Cut wrong-way dependencies ~73% with include-what-you-use, forward declarations, dependency inversion through startup-injected callbacks, and relocating misplaced code, in small reviewed changes to another team's code; in production. Presented the design in an org-wide tech talk.
- **Licensing for distributed proofs**: built token licensing in the C++ proof engine and fixed production license-checkout failures in the master/worker distributed solver.
- **Session save/restore**: extended snapshots to keep state that was silently lost on restore (user Tcl procedures, engine problem state as XML), and kept new features' state persistent with round-trip tests.

Software Engineer II | Microsoft *Noida, India | May 2021 - May 2023*

- Owned Recording Studio features in PowerPoint Win32 (C++): camera modes, live background blur, video export and Cameo, a camera object placed on the slide. Led v1 test automation and wired Recording and Export telemetry.
- Designed catalog filtering plus an allowlist (C#) so new commerce SKUs no longer broke OneDrive's upgrade path.
- Built the Stream 2.0 video playback component in Excel Win32 (C++, TypeScript) for SharePoint and OneDrive video.

Software Development Engineer 2 | Adobe *Noida, India | Aug 2015 - Apr 2021*

Software Engineer Aug 2015 - Jan 2017; SDE 2 from Jan 2017

- Moved Photoshop Elements and Premiere Elements from RIBS to Hyperdrive builds with parallel install bundles, cutting install time ~50% on average. Spot Award.
- Designed lazy loading for large files in the AEM web editor, cutting load time ~60% on files customers reported as freezing.

EDUCATION

B.Tech, Computer Science and Engineering - NIT Bhopal (MANIT) | 2011 - 2015 | GPA 8.27/10